

Assignment - 3

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Ans 1 @ We have, $\frac{\sigma}{\omega \epsilon} = \frac{8 \times 10^{-2}}{2\pi \times 10^6 \times 56 \times 3.6 \times 8.854 \times 10^{-12}} \approx 8$

Now $\alpha = \sqrt{\frac{\omega^2 \mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right]} \approx 5.410$ $\left[\begin{matrix} \omega = \pi \times 10^7 \\ \mu_r = 5.6 \\ \epsilon_r = 3.6 \end{matrix} \right]$

and $\beta = \sqrt{\frac{\omega^2 \mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} + 1 \right]} \approx 6.219$

$\therefore \gamma = \alpha + j\beta = 5.41 + j6.22$

⑥ $\lambda = \frac{2\pi}{\beta} = 1.025 \text{ m}$

⑦ Velocity (V) = $\frac{\omega}{\beta} = 5.125 \times 10^7 \text{ m/s}$

⑧ $\eta = \sqrt{\frac{j\omega \mu}{\sigma + j\omega \epsilon}} \Rightarrow |\eta| = \frac{1/\epsilon_0 \sqrt{\mu_r/\epsilon_r}}{\left[1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2 \right]^{1/4}} = 101.405$

$\angle \tan \delta = \frac{\sigma}{\omega \epsilon} = 8 \Rightarrow \delta = 86.86^\circ$ $\delta = 20^\circ$

$\therefore \angle \eta = 43.43^\circ$

$\Rightarrow \eta = |\eta| \angle \angle \eta = 101.405 \angle 43.43^\circ$

⑨ $\vec{H}_s = \frac{\hat{a}_x \times \vec{E}_s}{\eta} = \frac{-6}{\eta} e^{-\gamma z} \hat{a}_y = -59.17 e^{-j43.43^\circ - \gamma z} \hat{a}_y \text{ mA/m}$

Ans 2 @ $\eta = \frac{1}{\epsilon_0} \sqrt{\frac{\mu_r}{\epsilon_r}} \left[\sigma = 0 \text{ for lossless medium} \right] = 177.715 \angle 0^\circ$ $\left[\begin{matrix} \mu_r = 1 \\ \epsilon_r = 4.5 \end{matrix} \right]$

$\therefore \vec{H} = \frac{\hat{a}_z \times \vec{E}}{\eta} = 0.45 \cos(\omega t - \beta z) \hat{a}_y + 0.33 \sin(\omega t - \beta z) \hat{a}_x \text{ A/m}$

① We know $E_x = 8 \cos(\omega t - \beta z)$ & $E_y = -6 \sin(\omega t - \beta z)$
 In $z=0$ $E_x(0, t) = 8 \cos(\omega t)$ & $E_y = -6 \sin(\omega t)$

$$\therefore \frac{E_x^2}{8^2} + \frac{E_y^2}{6^2} = 1$$

As from above results, we can conclude that our eqn is of a left handed elliptically polarized.

② Now $\beta = \omega \sqrt{\mu \epsilon} = 1.77 \text{ rad/m}$

$$\lambda = \frac{2\pi}{\beta} = 3.535 \text{ m}$$

$$\mu = \frac{\omega}{\beta} = \frac{2\pi \times 40 \times 10^6}{1.77} = 1.41 \times 10^8 \text{ m/s}$$

Ans 3 We know, $\frac{\epsilon}{\epsilon_0} = 1.5$ [$\sigma = 4$, $\epsilon_r = 24$, $\mu_r = 1.6$]

$$\text{Now, } \alpha = \omega \sqrt{\frac{\mu \epsilon}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} + 1 \right] = 137.35 \text{ Np/m}$$

$$\text{Now, } 10^{-5} \epsilon_0 = 60 e^{-\alpha n}$$

$$\Rightarrow -5 \ln 10 = -\alpha n \Rightarrow n = 0.0838 \text{ m}$$

Ans 4 ① Given $\vec{E} = \cos(\omega t + \beta y) \hat{a}_x + \sin(\omega t + \beta y) \hat{a}_z \text{ V/m}$

In $y=0$ plane, $E_x(0, t) = \cos(\omega t)$

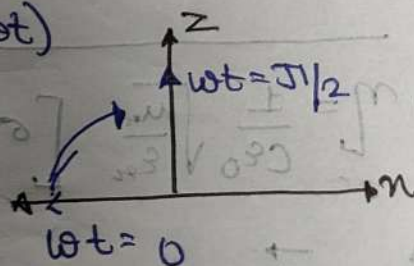
$$E_y(0, t) = \sin(\omega t)$$

$$\Rightarrow E_x^2 + E_y^2 = 1$$

and

$\therefore$ The eqn is of

left handed circular polarized.



$$\textcircled{b} \vec{E}_0 = 12e^{j\pi/3} e^{-j\omega t} \hat{a}_y + 5e^{-j\pi/3} e^{-j\omega t} \hat{a}_z \text{ V/m}$$

$$\Rightarrow \vec{E} = 12 \cos(\omega t - \pi/3) \hat{a}_y + 5 \cos(\omega t - \pi/3) \hat{a}_z$$

In $n=0$ plane, $E_y(0, t) = 12 \cos(\omega t + \pi/3)$

$$E_z(0, t) = 5 \cos(\omega t - \pi/3)$$

$$\Rightarrow E_z = 5 \cos(\omega t - \pi/3) = 5 \cos(\omega t + \pi/3 - 2\pi/3)$$

$$= 5 \left[\cos(\omega t + \pi/3) \cos(2\pi/3) + \sin(\omega t + \pi/3) \sin(2\pi/3) \right]$$

$\Rightarrow$ on observing, we get

$$(24 E_z^2 + 5 E_y)^2 = (5\sqrt{3})^2 + 5(144 - E_z^2) =$$

$$\Rightarrow \frac{4E_z^2}{75} + \frac{E_y^2}{108} + \frac{E_z \cdot E_y}{45} = 1$$

[The above eqn is an ellipse right handed polarized]

Ans 5 Let $\vec{E}_1 = E_0 [\cos(\omega t - \beta z) \hat{a}_x + \sin(\omega t - \beta z) \hat{a}_y]$ ①

$$\vec{E}_2 = E_0 [\cos(\omega t - \beta z) \hat{a}_x - \sin(\omega t - \beta z) \hat{a}_y]$$

As $\vec{E}_1$ is right handed & $\vec{E}_2$ is left handed circular polarized

$$\& \vec{E}_1 + \vec{E}_2 = 2E_0 \cos(\omega t - \beta z) \hat{a}_x \Rightarrow \text{Linearly polarized}$$

as $\vec{E}_y = 0$

Ans 6 Velocity $(v) = \frac{\omega}{\beta} \Rightarrow c\omega = c\beta = 12 \times 10^9 \text{ rad/s}$

$$\text{Now } \vec{E} = \frac{1}{\epsilon} \int \nabla \times \vec{H} dt$$

$$\nabla \times \vec{H} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & 10 \sin(\omega t - 40\pi z) \end{vmatrix}$$

$$= \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 0 & 0 & 10 \sin(\omega t - 40\pi z) \\ 0 & 0 & 0 \end{vmatrix}$$

$$= -800 \cos(\omega t - 40\pi z) \hat{a}_y - 400 \cos(\omega t - 40\pi z) \hat{a}_z$$

$$= -200 \cos(\omega t - 40\pi z) (\hat{a}_y + 2\hat{a}_z)$$

$$\therefore \vec{E} = -\frac{800}{\omega \epsilon} \sin(\omega t - 40\pi) \hat{a}_y - \frac{400}{\omega \epsilon} \sin(\omega t - 40\pi) \hat{a}_z$$

$$\approx -75399 \sin(\omega t - 40\pi) \hat{a}_y - 3770 \sin(\omega t - 40\pi) \hat{a}_z$$

$$\text{Now } \vec{P} = \vec{E} \times \vec{H} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 0 & 10 \sin(\omega t - 40\pi) & -20 \sin(\omega t - 40\pi) \\ -75399 \sin(\omega t - 40\pi) & -3770 \sin(\omega t - 40\pi) & 0 \end{vmatrix}$$

$$= [140780 \sin^2(\omega t - 40\pi) + 37700 \sin^2(\omega t - 40\pi)] \hat{a}_x$$

$$\therefore \vec{P}_{avg} = \frac{1}{2} [140780 + 37700] \hat{a}_x = 94240 \text{ W/m}^2$$

Ans 7: (a) $\omega = \beta c = 9 \times 10^8 \text{ rad/s}$ [$\beta = 3$]

(b) $\lambda_{air} = \frac{2\pi}{\beta} = \frac{2\pi}{3} = 2.094 \text{ m}$

(c) Loss Tangent ($\tan \delta$) = $\frac{\epsilon''}{\omega \epsilon'} = 2\pi = 6.288$ [$\epsilon'' = 4$, $\epsilon' = 80$]

Now, $\eta \Rightarrow \tan^{-1}\left(\frac{\epsilon''}{\omega \epsilon'}\right) = \theta = 20^\circ$

$\Rightarrow \theta = 40.478^\circ$

and $|\eta| = \frac{1}{c \epsilon_0} \sqrt{\frac{\mu_r}{\epsilon_r}} = 16.704$ [$\mu_r = 1$]

$$\left[1 + \left(\frac{\epsilon''}{\omega \epsilon'}\right)^2\right]^{1/4}$$

$\Rightarrow \eta = 16.704 \angle 40.478^\circ$

Now we know $\eta_{air} = 377 \Omega$

For a reflected wave, $\Gamma = \frac{\eta_{air} - \eta}{\eta_{air} + \eta} = \frac{377 - 16.704 \angle 40.478^\circ}{377 + 16.704 \angle 40.478^\circ}$

$$\Rightarrow \Gamma = 0.935 \angle 176.702^\circ$$

$$E_{reg} = \Gamma E_{inc} \quad (\text{in } +z \text{ direc})$$

$$\therefore \vec{E}_{reg} = 9.350 \sin(9 \times 10^8 t - 3z + 176.702^\circ) \hat{e}_m \text{ V/m}$$

Now, for transmitted wave

$$X = \omega \sqrt{\frac{\mu \epsilon}{2} [\sqrt{1 + (\epsilon/\omega \epsilon)^2} - 1]} = 43.955 \text{ Np/m}$$

$$\beta = \omega \sqrt{\frac{\mu \epsilon}{2} [\sqrt{1 + (\epsilon/\omega \epsilon)^2} + 1]} = 51.498 \text{ rad/m}$$

$$\text{Transmission coeff } T = \frac{2\eta_{oc}}{\eta_a + \eta_{oc}} = \frac{2 \times 16.704 \angle 40.478^\circ}{16.704 \angle 40.478^\circ + 377} = 0.0857 \angle 38.885^\circ$$

$$\vec{E}_{trans} = T \vec{E}_{inc} = 0.857 e^{43.955z} \sin(9 \times 10^8 t + 51.498z + 38.885^\circ) \hat{e}_x$$

Ans 8 From $\vec{E}_s \Rightarrow \vec{k} = -3.4 \hat{a}_x + 4.2 \hat{a}_y$

$$\therefore \vec{k} \cdot \vec{E} = 0 \Rightarrow -3.4 E_x + 4.2 E_y = 0 \Rightarrow E_x = 1.235 \text{ V/m}$$

$$\beta = |\vec{k}| = \sqrt{(-3.4)^2 + (4.2)^2} = 5.403 \text{ rad/m}$$

$$\text{Frequency } (f) = \frac{\beta c}{2\pi} = 258 \text{ MHz } [c = 3 \times 10^8 \text{ m/s}]$$

Now using $\nabla \times \vec{E}_s = -j\omega\mu\vec{H}_s$

$$\nabla \times \vec{E}_s = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1.235 e^{-j(3.4x - 4.2y)} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -j(3.4\mu - 4.2\mu) \\ -j(3.4\mu - 4.2\mu) & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$= e^{-j(3.4x - 4.2y)} [(12.6j - 16.8j)\hat{a}_x + (10.4j - 13.6j)\hat{a}_y - 8.54j]$$

$$\therefore \vec{H}_s = \frac{-0.491}{1000} e^{-j(3.4x - 4.2y)} [(12.8 + 16.8j)\hat{a}_x + (16.4 + 13.6j)\hat{a}_y + 8.54j]$$

Ans) (a) We have $\theta_{inc} = \tan^{-1} \left(\frac{1}{\sqrt{8}} \right) = 19.471^\circ$

We know $\theta_{ref} = \theta_i = 19.471^\circ$

Now using Snell's Law $n_1 \sin \theta_i = n_2 \sin \theta_t$

$$n_1 = \sqrt{\mu_{r1} \epsilon_{r1}} = 3 \text{ \& } n_2 = \sqrt{\mu_{r2} \epsilon_{r2}} = 1$$

$$\therefore 3 \sin(19.471^\circ) = \sin \theta_t$$

$$\Rightarrow \sin \theta_t = 90^\circ$$

(b) We know, $u = \frac{\omega}{\beta}$ & $u = \frac{c}{\sqrt{\mu_r \epsilon_r}} = 10^8 \text{ m/s}$

$$\therefore \beta = \frac{\omega}{u} = 10 \text{ rad/m}$$

Now, $\beta = k \sqrt{(1)^2 + (\sqrt{8})^2} \Rightarrow \frac{10}{3} = k \Rightarrow k = 3.33$

(c) The wavelength in dielectric, $\lambda_1 = \frac{u}{f} = \frac{10^8}{1.5 \times 10^8} = 0.628 \text{ m}$

wave $\lambda_{air} = \frac{3 \times 10^8}{10^9/2\pi} = 1.885 \text{ m}$

(d) $\vec{E}_{inc} = \eta_1 \vec{H}_i \times \hat{a}_{k_i}$ $\left[\eta_1 = \frac{1}{\epsilon_0} \sqrt{\mu_r/\epsilon_r} = 125.663 \text{ LO} \right]$

$$\therefore \vec{E}_{inc} = 125.663 \times 0.2 \cos(10^9 t - k_x x - k\sqrt{8}z) \hat{a}_y \times \left(\frac{1}{3} \hat{a}_x + \frac{\sqrt{8}}{3} \hat{a}_z \right)$$

$$= (23.695 \hat{a}_x - 8.377 \hat{a}_z) \cos(10^9 t - k_x x - k\sqrt{8}z)$$

© From Fresnel's eqn, we have

$$\Gamma_{11} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = \frac{377 \cos(90^\circ) - 125.663 \cos(19.47^\circ)}{377 \cos(90^\circ) + 125.663 \cos(19.47^\circ)} = -1$$

$$T_{11} = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = 6$$

$$\therefore \vec{E}_{\text{ref}} = T_{11} \vec{E}_{\text{in}} = (23.695 \hat{a}_x + 8.377 \hat{a}_z) \cos(10^9 t - kx - k\sqrt{8}z) \text{ V/m}$$

$$\vec{E}_{\text{tr}} = E_{i0} (-\cos \theta_t \hat{a}_x + \sin \theta_t \hat{a}_z) \cos(10^9 t - kx \sin \theta_t - k\sqrt{8}z \cos \theta_t) \text{ V/m}$$

$$E_{\text{tr}} = 6 E_{i0} = 6 \sqrt{(23.695)^2 + (-8.377)^2} = 150.793$$

$$\Rightarrow \vec{E}_{\text{tr}} = 150.793 \cos(10^9 t - kx) \hat{a}_z \text{ V/m}$$

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